EN3160: Image Processing and Machine Vision

Assignment 1: Intensity Transformations and Neighborhood Filtering

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GitHub Repo: https://github.com/RumethRanhinda/EN3160-Vision-UoM

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1. Intensity Transformation Implementation

[3]:
```python
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Verify library versions
print(f"OpenCV Version: {cv2.__version__}")
print(f"NumPy Version: {np.__version__}")

# Generate a 256x256 grayscale gradient to test rendering
gradient = np.tile(np.linspace(0, 255, 256, dtype=np.uint8), (256, 1))

# Render the test image
plt.figure(figsize=(5, 3))
plt.imshow(gradient, cmap='gray')
plt.title("Library Verification Successful")
plt.axis('off')
plt.show()
```

```
OpenCV Version: 4.11.0
NumPy Version: 2.1.1
```

Library Verification Successful